SONU KUMAR

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PROFESSIONAL SUMMARY:

AI/ML Engineer with hands-on experience building production-grade AI applications using FastAPI, Python, Large Language Models (LLMs), and modern backend architectures. Experienced in developing end-to-end AI products, including automated marketing content generation platforms, intelligent workflow automation, image generation pipelines, and scalable REST APIs. Skilled in integrating GenAI models, designing backend services, managing relational databases, optimizing AI workflows, and collaborating across frontend and backend teams to deliver enterprise software.

ACADEMICS QUALIFICATION:

MCA – GGSIP University (**9.4** CGPA) *Jul 2021 - Aug 2023*
BCA – PDM University (**6.3** CGPA) *Jul 2017 - Aug 2020*

COURSES & CERTIFICATES:

Advance Data Science Certification, Introduction to AI/ML, Introduction to Linux, Machine Learning & Algorithms.

TECHNICAL SKILLS :

Programming Language : PYTHON, JAVA, C, C++

AI/ML Frameworks : TensorFlow, PyTorch, Scikit-Learn, Langchain, NLTK, Keras, OpenCV, LangChain, Hugging-Face Transformers, SpaCy, SentenceTransformers, Faiss, Weaviate

GenAI & LLM Tools : LLMs (GPT, LLaMA, Mistral), RAG Pipelines, Prompt Engineering, VectorStores, ChromaDB, Gemini API

Backend : FastAPI, REST APIs, Async Programming, API Integration, Authentication, File Upload Handling, Image Processing Pipelines

Frontend : JavaScript, React.js, HTML, CSS

Data Analysis & Visualization : Pandas, NumPy, Matplotlib, Seaborn

DataBases & Tools : MySQL, SQLite, MS Office, Git & GitHub, MATLAB, Google Colab, Streamlit, Flask, FastAPI

AI/ML PRACTICAL EXPERIENCE SUMMARY :

- Developed **production-grade AI applications** using **Python**, **FastAPI**, **REST APIs**, and **Generative AI** for enterprise automation.
- Built **AI-powered marketing automation software** for automated promotional visual generation using **LLMs** and prompt engineering.
- Designed and implemented **RAG pipelines** using **LangChain**, **Gemini API**, **FAISS**, **ChromaDB**, **Pinecone**, and **Weaviate** for intelligent document retrieval.
- Integrated **LLMs (GPT, LLaMA, Mistral)** into AI applications for chatbots, document intelligence, and workflow automation.
- Engineered scalable **backend services**, **AI workflows**, and API integrations with **FastAPI**, **Flask**, and **Streamlit**.
- Designed and optimized **MySQL**, **SQLite**, and vector databases for structured, semantic, and embedding-based data retrieval.
- Built NLP solutions using **Hugging Face**, **SentenceTransformers**, **SpaCy**, and **NLTK** for semantic search, question answering, and text summarization.
- Developed machine learning and deep learning solutions using **TensorFlow**, **PyTorch**, **Scikit-learn**, and **Keras** for real-world computer vision and predictive analytics.
- Developed automation solutions, including **AI Voice Calling Agents**, intelligent workflows, and enterprise process automation using **n8n**.
- Built web scraping and data extraction pipelines using **Requests** and **BeautifulSoup** for automated data collection and AI-driven analysis.
- Utilized **Git**, **GitHub**, **Google Colab**, and modern development practices for collaborative software development, deployment, and version control.

WORK EXPERIENCE :

AI/ML Engineer at – Miracle Eye Pvt. Ltd. (*Nov 2025 -Present***)**

- Developed a **production-grade AI marketing automation platform** using **Python**, **FastAPI**, and **Generative AI** to automate promotional visual generation.
- Built scalable **FastAPI** backend services, **REST APIs**, and AI workflows for content generation, approval, and execution management.
- Designed and optimized **MySQL** databases, media management, secure file handling, and backend performance for enterprise applications.
- Integrated **LLMs**, **prompt engineering**, AI image generation, and **React/JavaScript** frontend with scalable backend services.

AI/ML Engineer Intern at – CyberSRC Pvt. Ltd. (*June 2025 -Nov 2025***)**

- Worked on large language models (LLMs) for custom NLP solutions, including fine-tuning and prompt engineering.
- Collaborated with teams to apply AI solutions for cybersecurity, including scraping exploit codes using requests and BeautifulSoup..
- Built and deployed conversational AI systems using LangChain and other frameworks.
- Developed intelligent chatbots integrated with external APIs and enterprise platforms.

Data Science Intern at - Unified Mentor

- Developed & deployed machine learning models for predictive analytics, with a focus on data-driven decision making
- Performed comprehensive data preprocessing, feature engineering, and model optimization.
- Implemented deep learning algorithms for high-accuracy image classification tasks.

IT-Recruiter at - Mbit Computraining Pvt. Ltd.

- Responsible for end-to-end recruitment in the Data Science and AI/ML domain, including sourcing, screening, and onboarding specialized talent.
- Conducted technical interviews and candidate assessments to evaluate skills and fit.
- Maintained candidate databases and streamlined recruitment workflows for improved efficiency.

PROJECT EXPERIENCE *(Company/Client Work):*

AI-Powered Marketing Automation Platform (Miracle Eye Pvt. Ltd.)

Tech Stack: *Python, FastAPI, Generative AI, LLMs, MySQL, JavaScript, React, REST APIs*

- Developed a **production-grade AI-powered marketing automation platform** using **Python, FastAPI, Generative AI, LLMs, and Prompt Engineering** to automate promotional visual generation through scalable AI workflows.
- Engineered **FastAPI REST APIs**, optimized **MySQL** database architecture, integrated **React/JavaScript** frontend modules, AI image generation pipelines, secure media management, and workflow automation for a scalable enterprise AI solution.

VOICE CALLING AGENT for Lead Automation

- Developed an automated voice calling workflow using n8n to engage leads, analyze call outcomes, schedule product demos, and send personalized follow-up emails. Integrated Excel-based analytics and multi-platform automation to streamline sales operations and improve lead conversion efficiency.

CWE & CVE Finder and COMPLIANCE Mapping Tool

- Built a RAG-based AI system leveraging vector stores and large language models to analyze company vulnerability reports and map them to relevant CWE and CVE entries. Utilized semantic search over large CWE CSV datasets and parsed CVE JSON archives for precise vulnerability identification. Additionally, mapped compliance control test cases according to the vulnerabilities and generated detailed impact descriptions using LLMs, enhancing threat analysis and regulatory compliance and generated detailed impact analyses through LLM-driven natural language generation, thereby improving the company's threat assessment accuracy and streamlining regulatory compliance workflows.

Exploit Code & Description Extraction

- Developed a web scraping tool using (Requests and BeautifulSoup) to extract exploit codes from ExploitDB and GitHub, enabling automated threat intelligence gathering and enrichment of vulnerability databases with real-world exploit references for improved security analysis Using GenAI to produce contextual summaries and descriptions, facilitating better understanding and faster response by security analysts.

Offline Personal Chatbot

- Made a fully offline conversational AI leveraging Mistral-7B (Q4 quantized) for language generation and E5-small-v2 embeddings, ensuring rapid and private interactions without reliance on internet connectivity. The architecture prioritized efficiency and data privacy, making it suitable for sensitive environments requiring autonomous AI assistance while maintaining low latency and high conversational quality.

PERSONAL PROJECT EXPERIENCE *(Independent or Open-Source Work):*

ATS Resume Matcher

- Built an AI-powered tool using LangChain and Gemini to evaluate CVs against job descriptions, providing ATS score-based matching percentages to improve resume alignment and job targeting.Enhanced recruitment efficiency by delivering precise resume-job fit analysis.

Real Mimic Messenger

- Designed a Hinglish conversational AI that mimics my unique messaging tone and informal chat patterns. Built to simulate real-time conversations with personalized emotional cues and dialect-specific phrasing. Enabled natural, personalized communication through culturally aware AI dialogue.

AI- Powered Blog Generator

- Built a full-stack AI tool that generates SEO-optimized blog articles based on user-provided titles by integrating web scraping, NLP, and content structuring. Enabled downloadable Word document export, automating content creation for bloggers and marketers. Automated content generation to improve productivity and SEO effectiveness.

Plant Leaf Disease Detection

- Developed a deep learning-based AI model for plant disease detection, achieving 94% accuracy, significantly enhancing early diagnosis and crop management. Provided actionable insights to aid early intervention and reduce crop loss.

AI Lawyer Chatbot

- Integrated a powerful (RAG) system using Gemini, LangChain, and Streamlit to develop an AI-powered legal assistant. Enabled intelligent interaction with PDF documents, providing accurate and context-aware answers to complex legal queries based on user-provided documents. Streamlined legal research by delivering precise, context-aware document analysis.

Face Mask Detection

- Programmed a YOLO-based AI model to detect face mask compliance in real-time, achieving 91% accuracy in classifying individuals as properly masked, unmasked, or improperly masked. Supported public health monitoring through real-time mask compliance detection.

ACADEMIC PROJECTS:

Online Examination System

- Implemented and deployed a secure, user-friendly online examination system, ensuring integrity, scalability, and a seamless user experience.

Personal Safe Web Browser

- Generated a custom web browser with enhanced security features, including a built-in incognito mode that runs by default to ensure private and secure web browsing.

Voice-Controlled Assistant

- Built a voice-activated assistant leveraging Natural Language Processing (NLP) to understand and respond to user commands. Integrated speech-to-text and text-to-speech capabilities for smooth and interactive communication.

ACHIEVEMENTS:

- Achieved an impressive **Rank of 248** in the **CET Entrance Exam** for MCA admission.
- Honored with the title of **Mr. Talented Fresher** Tag during college induction events.
- **Won 1st Prize in Solo Singing Competition** at the college cultural fest, demonstrating creative and musical talent
- Performed **Anchoring in multiple college events**, showcasing strong communication and public speaking skills.